

```
mongosh mongodb+srv://<credentials>@cluster0.hqljv0w.mongodb.net/
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Users\Sai Saahas\Downloads\mongosh-2.10.0-win32-x64\mongosh-2.10.0-win32-x64\bin

C:\Users\Sai Saahas\Downloads\mongosh-2.10.0-win32-x64\mongosh-2.10.0-win32-x64\bin>mongosh "mongodb+srv://cluster0.hqljv0w.mongodb.net/" --apiVersion 1 --username saahasva
ddi_db_user
Enter password: ****
Current Mongosh Log ID: 6a892a9aaa8496b7339b8447
Connecting to:      mongodb+srv://<credentials>@cluster0.hqljv0w.mongodb.net/?appName=mongosh+2.10.0
Using MongoDB:      8.0.29 (API Version 1)
Using Mongosh:      2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-gubp69-shard-0 [primary] test>

Atlas atlas-gubp69-shard-0 [primary] test> show databases;
sample_mflix 156.46 MiB
schoolDB      184.00 KiB
admin         0 B
local         0 B
Atlas atlas-gubp69-shard-0 [primary] test> create database bookflow_db;
Uncaught:
SyntaxError: Missing semicolon. (1:6)

> 1 | create database bookflow_db;
    |      ^
  2 |

Atlas atlas-gubp69-shard-0 [primary] test> show databases;
sample_mflix 156.46 MiB
schoolDB      184.00 KiB
admin         0 B
local         0 B
Atlas atlas-gubp69-shard-0 [primary] test> use BookFlowDB
| db.createCollection("book_metadata")
switched to db BookFlowDB
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.insertMany([
| {
```

```

mongosh mongodb+srv://<credentials>@cluster0.hqjv0w.mongodb.net/
switched to db BookFlowDB
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.insertMany([
{
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: { pages: 340, fileSize_MB: 4.2, drm: true, narrator: "K. Rao", runtime_min: 610 },
  avgRating: 4.6,
  reviews: [
    { member_id: "M1001", rating: 5, comment: "Excellent deep dive into digital systems design.", date: ISODate("2023-01-15") },
    { member_id: "M1042", rating: 4, comment: "Good but the Digital edition had formatting issues.", date: ISODate("2023-02-20") }
  ]
},
{
  title: "Ocean of Code",
  author: "A. Fernandes",
  genre: "Technology",
  publishedYear: 2021,
  formats: ["Digital", "Paperback"],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    { member_id: "M2001", rating: 4, comment: "Solid Digital read, clean layout.", date: ISODate("2022-05-10") },
    { member_id: "M2002", rating: 5, comment: "Best programming book this year.", date: ISODate("2022-06-01") }
  ]
},
{
  title: "Whispers of Yesterday",
  author: "S. Kapoor",
  genre: "Fiction",
  publishedYear: 2019,
  formats: ["Hardcover", "Paperback"],
  specs: { pages: 410, binding: "Sewn" },
  avgRating: 3.8,
  reviews: [
    { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
    { member_id: "M3002", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
  ]
},
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 },

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  title: "Whispers of Yesterday",
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  specs: { pages: 410, binding: "Sewn" },
  avgRating: 3.8,
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    { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
    { member_id: "M3002", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
  ]
},
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    { member_id: "M4001", rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") },
    { member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.", date: ISODate("2023-09-14") }
  ]
}
])
})
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a892df6aa8496b7339b8448'),
    '1': ObjectId('6a892df6aa8496b7339b8449'),
    '2': ObjectId('6a892df6aa8496b7339b844a'),
    '3': ObjectId('6a892df6aa8496b7339b844b')
  }
}
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.find({ avgRating: { $gte: 4 } }, { title: 1, avgRating: 1, _id: 0 })
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } }, { title: 1, formats: 1, _id: 0 })
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Digital', 'Audiobook' ]
  }
]
```

```
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```

```
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.find({ avgRating: { $g$gt: 4 } }, { title: 1, avgRating: 1, _id: 0 })
```

```
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
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Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } }, { title: 1, formats: 1, _id: 0 })
```

```
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]
```

```
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.find({ avgRating: { $g$gt: 4 }, formats: { $in: ["Digital"] } })
```

```
[
  {
    _id: ObjectId('6a892df6aa8496b7339b8448'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  }
]
```

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    date: ISODate('2023-02-20T00:00:00.000Z')
  }
],
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  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a892df6aa8496b7339b844b'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.'
```

```

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  date: ISODate('2022-06-01T00:00:00.000Z')
}
]
},
{
  _id: ObjectId('6a892df6aa8496b7339b844b'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]
}
]
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.find({ formats: { $reg$regex: "Digital", $options: "i" } }, { title: 1, formats: 1 })
[
  {
    _id: ObjectId('6a892df6aa8496b7339b8448'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a892df6aa8496b7339b8449'),
    title: 'Ocean of Code',
    formats: [ 'Digital', 'Paperback' ]
  },
  {
    _id: ObjectId('6a892df6aa8496b7339b844b'),
    title: 'Quantum Horizons',
    formats: [ 'Digital', 'Audiobook' ]
  }
]

```

```

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[
  {
    _id: ObjectId('6a892df6aa8496b7339b8448'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a892df6aa8496b7339b8449'),
    title: 'Ocean of Code',
    formats: [ 'Digital', 'Paperback' ]
  },
  {
    _id: ObjectId('6a892df6aa8496b7339b844b'),
    title: 'Quantum Horizons',
    formats: [ 'Digital', 'Audiobook' ]
  }
]
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" } }, { title: 1, "reviews.$": 1 })
[
  {
    _id: ObjectId('6a892df6aa8496b7339b8448'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a892df6aa8496b7339b8449'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a892df6aa8496b7339b844b'),
    title: 'Quantum Horizons',
    reviews: [

```



```

]
}
]
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.aggregate([
{ $match: { publishedYear: { $gt: 2020 } } },
{ $unwind: "$reviews" },
{ $group: {
_id: "$title",
avgReviewRating: { $avg: "$reviews.rating" },
totalReviews: { $sum: 1 },
author: { $first: "$author" }
}},
{ $sort: { avgReviewRating: -1 } },
{ $project: {
_id: 0,
title: "$_id",
author: 1,
avgReviewRating: { $round: ["$avgReviewRating", 2] },
totalReviews: 1
}}
])
[
{
totalReviews: 2,
author: 'R. Menon',
title: 'Quantum Horizons',
avgReviewRating: 5
},
{
totalReviews: 2,
author: 'A. Fernandes',
title: 'Ocean of Code',
avgReviewRating: 4.5
},
{
totalReviews: 2,
author: 'R. Menon',
title: 'The Silent Algorithm',
avgReviewRating: 4.5
}
]
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.createIndex({ "reviews.comment": "text" })

db.book_metadata.find(
{ $text: { $search: "Digital" } },
{ title: 1, score: { $meta: "textScore" } }
).sort({ score: { $meta: "textScore" } })

```



```

mongosh mongodb+srv://<credentials>@cluster0.hqjiv0w.mongodb.net/
}
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB> db.book_metadata.createIndex({ "reviews.comment": "text" })

db.book_metadata.find(
{ $text: { $search: "Digital" } },
{ title: 1, score: { $meta: "textScore" } }
).sort({ score: { $meta: "textScore" } })

db.book_metadata.find({ $text: { $search: "Digital" } }).explain("executionStats")

db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" } }).explain("executionStats")

db.book_metadata.createIndex({ format: 1 })
db.book_metadata.createIndex({ avgRating: -1 })
db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlowDB.book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: '85BA8276',
    planCacheShapeHash: '85BA8276',
    planCacheKey: '6EB4A3FD',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { avgRating: -1 },
        indexName: 'avgRating_-1',
        isMultiKey: false,
        multiKeyPaths: { avgRating: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { avgRating: [ '[inf.0, 4)' ] }
      }
    },
    rejectedPlans: []
  }
}

```

mongosh mongodb+srv://<credentials>@cluster0.hqjiv0w.mongodb.net/

```
    indexVersion: 2,
    direction: 'forward',
    indexBounds: { avgRating: [ '[inf.0, 4)' ] }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 1,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      keyPattern: { avgRating: -1 },
      indexName: 'avgRating_-1',
      isMultiKey: false,
      multiKeyPaths: { avgRating: [ ] },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: { avgRating: [ '[inf.0, 4)' ] },
```

```
mongosh mongodb+srv://<credentials>@cluster0.hqljv0w.mongodb.net/
  dupsDropped: 0
}
}
},
queryShapeHash: '32937274CD04FCE30803C45B066E908C738D8F34F4836BD1BDD8198606AEBC7F',
command: {
  find: 'book_metadata',
  filter: { avgRating: { '$gt': 4 } },
  '$db': 'BookFlowDB'
},
serverInfo: {
  host: 'ac-gl5cwpm-shard-00-02.hqljv0w.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787375269, i: 35 }),
  signature: {
    hash: Binary.createFromBase64('py0hCnVRL7y/oGm7Vvq0ULWpOC0=', 0),
    keyId: Long('7634960356174987267')
  }
},
operationTime: Timestamp({ t: 1787375269, i: 35 })
}
Atlas atlas-gubp69-shard-0 [primary] BookFlowDB>
```